

Communication with Individuals with Intellectual Disability Using Generic Images: Exploring Contexts, Requirements, Opportunities, and Challenges

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Abstract

This paper examines the use of generic images depicting natural scenes, combined with browsing technologies, in non-verbal interactions with individuals with intellectual disabilities. Current Augmentative and Alternative Communication (AAC) methods often demand extensive learning and memorization of symbolic visuals, creating usability challenges. This case study report examines the potential for adapting the Canvis app, an image-based communication prototype, in real-life scenarios for adults with intellectual disabilities. Through this case study, we identify user challenges and provide design recommendations for improvement. The case study presents observations made during a three-month internship at two community hubs in Queensland, Australia. We highlight that effective communication technology for this group of users must encompass a broader perspective, prioritizing connection, self-expression, and participation rather than merely symbol exchange. The findings suggest that generic image browsing platforms have the potential to enhance communication experiences for users, fostering inclusivity and enabling meaningful engagement with others.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; • **Social and professional topics** → **People with disability**.

Keywords

Image-Based Communication, Non-verbal Communication, Intellectual Disability, Generic Image Browsers

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1 Introduction and Background

Individuals with intellectual disability often rely on non-verbal and non-linguistic methods to communicate with their support network. Functional communication [9] skills are crucial for social inclusion and community engagement. However, leading conversation topics is often challenging for this group, and as a result, their interactions are typically guided by supporters and carers [18].

Assistive communication technologies, such as Picture Exchange Communication Systems (PECS) [5, 15], Augmentative and Alternative Communication (AAC) [21, 25], and Visual Scene Displays (VSDs) [2], have been suggested to improve communication skills for a broad range of users with language impairment, cognitive disability and intellectual disability. Nonetheless, there remain challenges in their utilization and effectiveness for adults with intellectual disability. These challenges include limited support to express abstract concepts, match symbol representations with their conceptualization, and search for vocabulary, especially in novel contexts [6, 14, 18].

This limitation underscores the necessity for more adaptable and intuitive communication aids that can effectively support dynamic real-life interactions. Wilson et.al [27] co-designed "Put Yourself in the Picture", a mobile app aimed at empowering young adults with intellectual disability to visualize themselves by augmenting images with depictions of themselves, achieving personal goals. The app was utilized for social activities and supported individuals beyond the planning process, contributing more broadly to enhancing overall communication, self-expression, and socialization. Exploring the potential of Optical Character Recognition (OCR) to enable image-rich real-life environments, the Emoji Assistant mobile application [22] was designed to offer visual interpretations of textual restaurant menus, to enhance communication and collaboration among users with intellectual disability and their support workers during the meal ordering process.

Robertson et al. [18] previously co-designed the Canvis application to assist people with verbal disability in visually expressing their thoughts through grid-based image albums. This research investigated the use of generic images in communication that encompass natural scenes, meaningful events, as well as various subjects like people, animals, and objects. Generic images, also employed by the new wave of visual communication tools such as Visual Scene Displays (VSDs) [2, 11], are easily comprehensible and provide a rich context for conversations [16, 26]. The Canvis application and

its image-exploring process showed promise in enriching, supporting, and orienting the conversations by offering access to a diverse array of visuals without relying on linguistic prompts [23].

In this paper, we document the opportunities and challenges for embedding image-based communication in a range of daily activities through a case study drawn from a three-month internship in two learning and lifestyle centers of a disability service provider. The intern, a computer science research student with a year of experience in co-designing technology for individuals with disability, explored various contexts in which generic images and image browsers could be used as communication aids. The organization's mentors sought to identify opportunities and challenges in adopting and integrating the Canvis app into daily communications and to inform future research for enhancing its interface and algorithms to add value to the support centers.

By sharing and discussing the opportunities and challenges we observed, we hope to contribute to a new wave of person-centered visual communication approaches that leverage everyday pictures throughout everyday activities. Researchers developing technology for individuals with intellectual disability will find inspiration in this report for designing more inclusive communication tools that offer greater choice and address some challenges for sharing and connecting for this group. Disability organizations, caregivers, and support personnel will discover new approaches to leveraging technology for enhanced support for their clients.

2 Context and Setting

2.1 Disability Support Centers

The case study presented involves people with intellectual disability who are supported by a Disability Support Organization (DSO) in Queensland, Australia. This DSO assists their clients with intellectual disability in socializing, accessing community resources, and learning daily living skills. Intellectual disability is defined by the Australian National Disability Insurance Scheme (NDIS) in line with DSM-5 criteria [1], which are divided into deficits in intellectual functioning (including reasoning, problem-solving, planning, abstract thinking, judgment, academic learning, and experiential learning) and adaptive functioning (including communication, social skills, personal independence, and school or work functioning).

In this paper, we do not engage with the clinical diagnoses of our clients in relation to their use of image-based communication technology. As proposed by [24], we instead use an individualized approach that focuses on the support required for daily activities—rather than specific diagnostic labels—as this provides a more accurate reflection of a person's cognitive abilities and their interaction with technology. This support-based approach is particularly beneficial since not all individuals with intellectual disability receive a formal medical diagnosis. By focusing on support needs, we can more effectively assess how the technology meets the diverse needs of its users.

The case study reflects on experience in 2 community hubs, which operate as day centres. Each center provides assistance to approximately 30 adults with varying levels of verbal abilities and diverse support needs. Some individuals receive personalized support within the facility, while others participate in group activities with a 1:4 ratio of support worker to clients. Support workers facilitate

a range of group-based lifestyle activities such as communication practice, shopping, cooking, 3D printing, art, English, self-care, environmental management, and robotics. Computers and tablets are available for those interested, and clients are exposed to online information daily. Although some clients have access Augmentative and Alternative Communication (AAC) applications on their tablets, these tools are infrequently used for communication.

2.2 Tools and Applications

The Canvis application, which we obtained from its authors [18], prompted the exploration and this case study report. However, other digital image collections, such as Google image search or personal images on the center's devices were also included. Canvis allows access to pictures in three ways: a) from Flickr through textual search and pre-defined categories (see Figure 3), b) through personal photos stored on the device and c) through an interactive grid of images from the MS-COCO dataset (see Figure 2). The grid is initialised with 180 random images and every time an image is tapped on this grid, the grid is refreshed with a set of 35 images similar to the one tapped, assisting users in iteratively seeking desired images. The latest tapped image features on the right-hand side of the screen with a validation button that enables users to place it on their blank vision board (See Figure 1). The primary objective behind designing this image browser was to allow users to access a broad array of topics and images without relying on verbal prompts.

2.3 Positionality and internship's role and responsibility

As a computer science research student with about a year of experience in co-designing technology with individuals with disability, the first author completed a three-month internship with the DSO. She is an international student whose native language is not English. She had been living in Australia for less than a year when the internship began. Her background is in computer science, and her prior personal experience of establishing rapport with adults with intellectual disability had been limited to previous co-design sessions. She additionally has personal experience in caring for young children. Both her mentors, the second and third author of this paper, have over 10 years of experience in innovative technology co-design with adults with intellectual disability in Australia, primarily from a research and implementation perspective respectively.

The internship focused on exploring the practical application Canvis, an application presented as part of a research project involving generic images for communication and self-expression [18]. The first author's tasks included:

- (1) Exploring everyday communication scenarios where the Canvis app could be integrated.
- (2) Identifying environmental and technical challenges faced by a diverse range of users and their carers in implementing the App.
- (3) Providing suggestions for an improved version of the Canvis' interface and algorithms so that the next app can better support daily conversations.

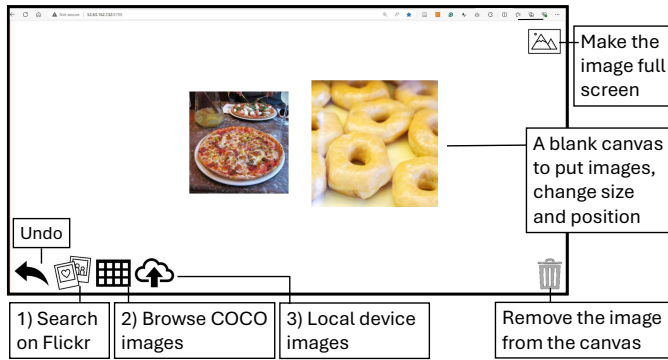


Figure 1: Canvis app-Blank canvas with image access options on the bottom left

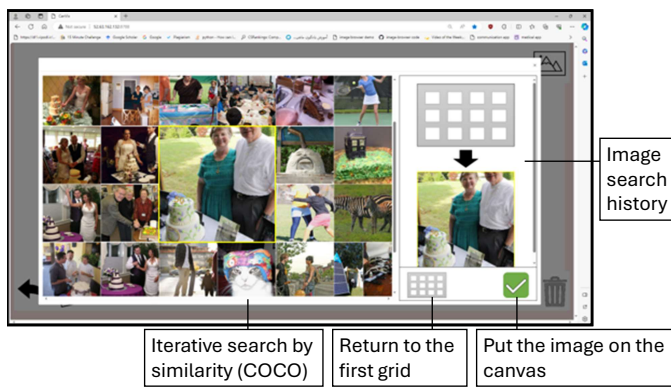


Figure 2: Canvis app-Image grid for iterative image search

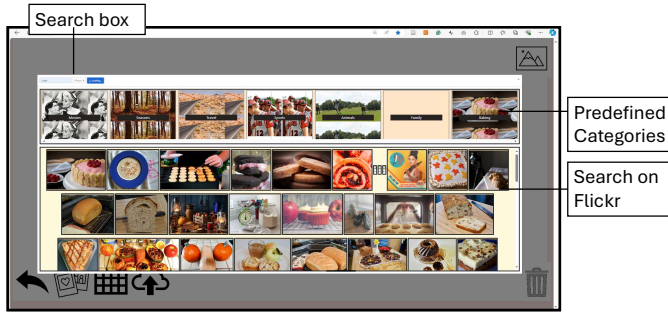


Figure 3: Canvis app-Image search and pre-defined categories

During the internship, she documented her observations from participating in various activities at the centers, both indoor and outdoor, five days a week. Based on the activity schedules of each center, their similarities and differences, and the feasibility of using images, the intern decided when to be present at each location. On average, she worked three to four days a week at the first center and one to two days a week at the second center. The first center supported a greater variety of structured group activities each for a specific number of individuals (5-6), while the second center offered a more open environment and greater flexibility in participant engagement. The observations were compiled into detailed internal

reports reflecting the dynamics of communication in these settings. She also discussed her findings regularly with organizational and academic mentors (the second and third authors of this report) in weekly meetings.

Introducing the Canvis app [18] to clients aimed to foster meaningful connections, and observing their interactions with the app in natural settings was a key focus. Discussions with families, support workers, and speech pathologists provided valuable insights into the app and other communication strategies, which were crucial for refining the approach.

2.4 Communication Activities

In addition to observing activities and identifying communication contexts, the internship involved conducting various image-based activities within the centers to explore how generic images and browsers could enhance communication. Examples of these activities include:

- Utilizing images to prompt discussion about people's favorite topics: The objective was to establish conversational contexts based on their interests, facilitating communication and fostering friendships.
- Engaging individuals in sharing their experiences, memories, and emotions related to activities such as cooking, playing, and crafting through personal images that were available on the center's devices.
- Implementing enjoyable, game-like activities. For instance, clients were tasked with finding specific items within the image collection, allowing us to observe their browsing patterns. In another activity, clients were prompted to think of something and choose an image, while the group guessed their intentions. This exercise shed light on how individuals express intentions through images and interpret visual content.
- Looking for the opportunity to deploy the app in group activities and integrate images into class discussions or group planning sessions: in activities such as communication classes, café outings, and cooking sessions, individuals were prompted to discuss topics like self-care, emotion regulation, and café planning using prepared images.
- Soliciting preferences regarding the type of images or videos in different contexts: clients were asked whether they preferred animated or natural visuals, tailoring the communication aids to their preferences.

3 Reflections, Challenges, and Suggestions

In this section, we provide a reflection on the outcomes of the exploration during the three months of implementation of image-based activities, and we elaborate on potential contexts for this type of communication and its challenges. This reflection is written in the first person, from the perspective of the first author. Support centers for people with intellectual disability were rich environments including cooperative activities with support workers who make sure clients feel valuable, influential, and confident while engaging in activities, learning life skills, and accepting responsibilities.

3.1 Generic Image Browsers for Building Interpersonal Connections

Generic images had a remarkable ability to foster meaningful conversations and interpersonal connections. These images, rich in context, naturally prompted discussions about events, personal experiences, and interests. Individuals with intellectual disability, including those who are verbal, often struggle with abstract thinking, generating topics, and initiating conversations. Unlike AAC systems, which often require constructing semi-structured sentences, recalling symbols, and navigating complex categories—processes that can feel cumbersome and disconnected from spontaneous conversation—generic images offer a powerful alternative by conveying substantial information without relying on words.

Clients using Canvis to browse images were not confined to predefined starting points and could explore a wide array of ideas, which facilitated greater engagement and self-expression. For instance, one person eagerly shared the joy of her father's birthday party by using an image of a celebration to initiate the conversation. This image served as both a memory prompt and a direct link to the shared experience, sparking relevant questions from others and effortlessly deepening the connection. Subsequently, many individuals at the centers welcomed the opportunity to view generic images, frequently approaching me to see photos and quickly familiarizing themselves with me as the intern. Even those initially disinterested eventually warmed to the images, which was particularly helpful for me as someone coming from a different background and needing time to build connections.

The ability to explore a variety of topics through a wide array of images, without needing verbal prompts, served as a catalyst for recalling untold stories, interests, memories, and ambitions that had previously gone unspoken. People were eager to share their interests through visual boards, which they used to communicate with others. Figure 4 illustrates two examples of these boards, created during discussions about their interests.

One woman expressed a keen interest in searching for images, both by browsing and typing keywords, to convey her preferences without verbal explanations. This allowed me to better understand her. Through these interactions—observing her search process and image selection while she responded to my questions with 'yes/no' answers—I learned about her interests and personal details, such as the loss of her dog, which she had never shared verbally. This process not only revealed more about her but also strengthened our emotional bond, leading to frequent invitations to participate in activities together. Support staff noted that, although she can speak and type, she typically engages in minimal verbal communication and finds social interactions challenging.

In addition to initiating conversations, the suggestion of new or random images within Canvis often sustained and deepened the conversation by introducing new topics or perspectives. For example, a client selected a photo of a favorite animal, leading to discussions about relevant activities, memories, or food, thereby strengthening the relationship over time.

These reflections highlight the opportunity for open-ended visual communication tools to support changes, such as new caregivers, friendship groups, teachers, or healthcare providers. Canvis offered a unique opportunity to enhance connections and build new

friendships through self-expression and sharing interests, especially with those being met for the first time. Disability services and caregivers can use such a tool to prepare and facilitate communication with new people.

3.2 Generic images for enhancing engagement in group activities

Support centers offer diverse multi-modal educational content facilitated by one or two support workers, catering to various groups of individuals with around 4 to 6 clients. These activities cover a range of topics such as cooking, communication, environmental awareness, self-care, household management, and shopping. Clients usually are encouraged to answer questions or participate in tasks such as sorting, categorizing, and planning items. While these activities are designed to be interactive and stimulating, some clients struggled to fully engage, particularly those with limited verbal abilities.

Clients' levels of engagement often correlate with the support worker's personality, creativity, and teaching skills. To address this issue, support workers occasionally utilize generic images as aids to enhance engagement. These images serve as visual prompts during discussions and activities, aiding individuals to participate more actively. For example, while co-facilitating an activity with a support worker, I showed images on the touchable large screen during discussions about emotion regulation or planning a visit to a café. This visual approach engaged people by prompting questions about their awareness of different emotions, their favorite type of café, preferred drinks, and related topics such as how to make a café, and what tools are needed. Figure 5 shows examples of images used in these group discussions.

Incorporating relevant visuals into real-time discussions enhanced engagement, sustained dialogue, and created a more inclusive environment for all clients, regardless of their verbal abilities. Relevant visuals can significantly boost conversation, but such images may not always be readily available without prior searching or preparation by support workers. Image browsers have significant potential to improve image access and group activity engagement if they can retrieve images based on semantic similarity and relevant concepts. For instance, during a discussion about cooking a specific dish, having immediate access to visuals of ingredients, tools, and cooking activities can be highly beneficial.

Additionally, when people were asked to search for specific items in an image collection, I paid attention to several items that clients struggled to find. I identified relationships and image associations other than visual similarity. Examples of categorical similarity, part-to-whole relationships, and conceptual similarity between selected images and their target items were evident in my findings. For example, to locate an animal, participants might click on images of other animals within the same category (e.g., birds or reptiles). Similarly, to find tomatoes, they might select images of pizza, or to find bread, they might refer to images of donuts.

We suggest that improving image retrieval algorithms to accommodate these types of associations and predict conversational intents based on the sequence of image selections could enhance communication around specific concepts in the future. Moreover, to utilize images in group activities, we suggest designing default



Figure 4: Visual boards made by clients for self-expression



Figure 5: Sample set of images used for group discussions

backdrops in an app like Canvis to facilitate sorting, linking, rating, listing, and categorizing various images during conversations. This will help support workers and other audience prompting the clients to answer their questions visually and actively engage in the activities. The possibility of adding shapes, text, and drawings inside the app will create more potential for conversation around various topics.

3.3 Exploratory Image Browsing for Leisure and Entertainment

Many clients exhibited a preference for leisurely activities over structured tasks, often gravitating toward the exploration of image browsers. Engaging in scrolling through these platforms, they delighted in discovering and admiring their favorite images, which can range from food and animals to various places and events. This recreational pursuit became a source of enjoyment for many, akin to a leisurely pastime.

However, while image browsing proved to be a source of entertainment, it occasionally led to a diversion from the original task or purpose. Clients became at times engrossed in exploring images of diverse topics, straying from their initial intent.

Frustrations often arose when individuals could not find the specific images they were looking for, especially when using the Canvis app where the desired image was not found in the initial grid. As a result, I had to rely on textual searches using Flickr or Google Image Search and their suggestions as alternative tools to locate users' target concepts. This experience highlighted the need for improved image retrieval algorithms that simplify the process of finding desired images with fewer steps. Additionally, recognizing

that users have diverse interests and needs, incorporating user profiles within the system could facilitate the ranking, personalization and customization of images based on individual priorities.

3.4 Visual Aids for Planning and Reporting

Through my discussions within the support network, as mentioned in section 3.1, it has become evident that individuals with intellectual disability greatly benefit from understanding their daily and weekly schedules and being prepared for unforeseen circumstances, such as navigating unfamiliar environments or encountering changes in caregivers or teachers. Visual prompts play a crucial role in facilitating their comprehension and planning of various tasks and routines. For instance, when planning outings to places like cafes, engaging individuals in discussions about the venue, menu options, and the people they might encounter can help ascertain their preferences and anticipate and mitigate unexpected reactions.

Furthermore, I have observed a strong interest among some people in reminiscing about past activities, such as music classes, bowling outings, or engaging in craft-making, using personal images on the centers' devices or personal albums. This interest was particularly noticeable during activities like scrapbooking. The option to use a grid of generic images alongside uploading personal photos was helpful for continuing conversations about activities and events. For example, I asked an individual to select pictures of themselves from an album, place them on a canvas, and describe what they were cooking at that moment or how they were feeling. To prompt a response, I opened the generic image browser, and they pointed to a picture of a pizza.

Moreover, some individuals demonstrated the capability to independently document their progress through self-written reports. These individuals are mostly those whom the centers want to support in taking on certain responsibilities, such as conducting training sessions for other clients, providing peer support, or performing tasks for the community. However, some of them lack the skills and confidence needed for report writing, often because they are unsure what topics to address in their reports. The centers are keen to help them improve in this area. The goal is for them to better reflect on their own and others' activities, express their feelings, and articulate the skills they have learned or wish to learn.

Based on these observations, we suggest that including relevant images of daily activities alongside personal photos can provide

essential contextual support and improve the usability of the Canvis app. Additionally, incorporating a feature that recommends semantically related images (based on the similarities discussed in Section 3.1) for personal photos could further facilitate discussions by acting as prompts to recall various topics for inclusion in reports. For instance, consider an individual who wants to reflect on how they supported a friend in cooking. With access to images related to different types of food, kitchen tools, utensils, and associated activities—along with algorithms that enable quick retrieval of these images—the individual would have a wealth of visual cues to help them write about the activity and explore various aspects of the topic.

4 Discussion

4.1 Enhancing communication

Based on the observations and reflections, we argue that generic image browsers provide a wide range of meanings and applications that help initiate and sustain conversations, thereby fostering quicker connections and the formation of friendships. This stands in contrast to AAC, VSD, and symbolic approaches, which are proposed to be effective for language development through predefined symbols and depend on recalling words to maintain a conversation [8, 12, 17, 20]. The naturalistic aspect of generic image shares similar benefits to naturalistic scenes in VSDs for language learning and communication, which have been shown to increase communication frequency and expand vocabulary [4, 14, 16] for individuals with complex needs. Generic images can maintain functional and visual relationships akin to real-world experiences, requiring less cognitive effort compared to symbolic grid displays [26]. They take advantage of humans' ability to rapidly process visual scenes, allowing for quick comprehension.

Thus, we emphasize the need for a more holistic adoption of assistive communication technologies for individuals with intellectual disability—one that extends beyond facilitating the exchange of words or symbolic visuals. Effective communication encompasses a broader spectrum of connection, self-expression, and participation including the conveyance of interests, thoughts, and emotions, which should not rely solely on verbal descriptions. Generic images can complement AAC and symbolic approaches by offering additional benefits of flexibility for other aspects of communication, which often need a deeper understanding than words alone [19].

4.2 Reducing Participation Barriers

We highlighted the potential of leveraging generic image browsers to increase engagement and participation in group activities. This aligns with research where visual objects and images are commonly used as projective elicitation, or priming tools, enabling individuals to express their ideas independently, facilitating discussions on sensitive topics, and activating subconscious schemas [7]. Visual methods in communication promote dialogue through visual thinking strategies, fostering awareness, empathy, and understanding [3, 7, 10]. Research on enhancing functional communication for adults with intellectual disability, emphasizes effective, accessible, and assistive communication systems, combined with conducive physical environments and trained supporters or carers [6, 9, 14]. In this regard, we argue that generic image browsers can help alleviate

community engagement barriers caused by physical or environmental factors. Consistently incorporating this model of communication technology within disability support services could significantly enhance the lives of individuals with intellectual disabilities by fostering more inclusive and adaptable communication and promoting greater involvement in their communities and co-design processes where their input is invaluable.

4.3 Improving Social Skills

We propose that unlike query-based platforms, a browsing-based platform like Canvis [18] can be an effective tool for entertainment or planning future activities by allowing users to recall new topics of interest. Similarly, proposing contextual images for their personal albums will prompt people to think about their past experiences from multiple perspectives and reflect on their progress notes. The effectiveness of the app in this area depends on including collections of images relevant to users' lives, as well as personalized, intelligent, and semantic-based image retrieval algorithms that align with users' intents [13] and browsing preferences. These features facilitate easier access to desired images and ultimately enhance the usability of the app in developing necessary skills for different clients.

4.4 Limitations and Future Works

This paper does not present a formal research study; the observations reported here are not direct quotes from the internship notes, nor were these notes analyzed thematically within a systematic framework. Instead, using a case study approach, the authors identified and agreed upon key themes and anonymized examples of interest that emerged during discussions throughout and at the conclusion of the internship. Insights from informal conversations with support workers and clients' families were also incorporated into the findings and discussion. Given the limited scope of this study, future research should broaden its focus to include other populations with intellectual disabilities and diverse cultural contexts. Plans for future work include enhancing the app with advanced image retrieval algorithms to increase image accessibility and evaluating its efficacy through quantitative metrics and structured user feedback.

5 Conclusion

This case study report explored the benefits and potential applications of using generic images in various communication contexts with individuals with intellectual disability, drawing from a three-month internship at two disability support centers in Queensland, Australia. We advocate for the integration of generic image browsers in assistive communication technologies and their diverse deployment within disability organizations to support broader aspects of human communication beyond verbal skills, including self-expression, social connection, and active participation. Our findings highlighted several technical challenges, such as enhancing the relevance of images to users' daily activities and improving image accessibility through semantic retrieval algorithms and intent recognition. Addressing these challenges presents valuable research opportunities to develop more inclusive and effective communication technologies.

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